The Effects of $\vec{B}$

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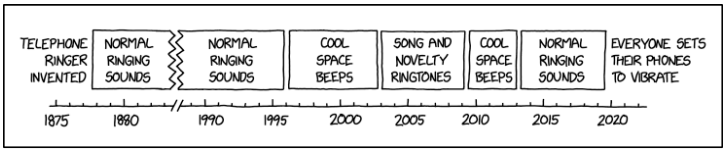

AFTER 140 YEARS, HUMANITY IS FINALLY ON THE VERGE OF WINNING THE WAR AGAINST RINGTONES.

1 Ferromagnets & Permanant Magnets

Atoms in a ferromagnet (iron, nickel, cobalt, etc) behave as if they were little current loops.

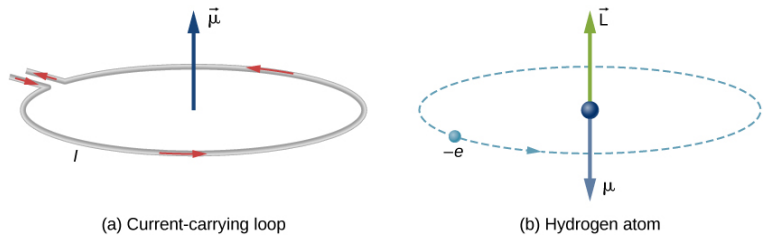

Figure 1: Atom compared to a current loop

Normally, these $\vec{B}$ fields point in all sorts of directions, due to the random movements of atoms. However, if a magnet was brought close, the atoms will align to a direction.

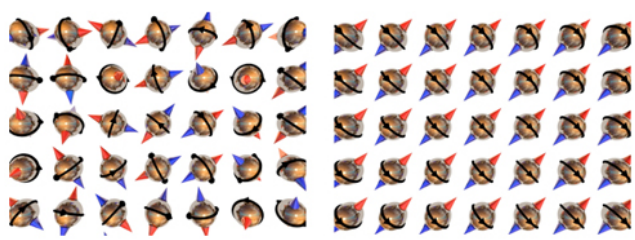

Figure 2: Magnetized atoms

Usually when the magnet is removed, the magnetization effect on the atoms vanishes. Some specialized alloys, however, such as steel, retain the magnetization and become permanant magnets. This also changes with temperature (lower temperature decreases Brownian motion).

2 The Lorentz Force Law

No magnetic equivalent of electical charge, so no basic equivalent of $F = qE$ or $F = mg$. Instead, we have the Lorentz force law:

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\vec{F} = q\vec{v} \times \vec{B}$$

- $\vec{v}$ is velocity of charge being forced
- $\vec{B}$ is the magnetic field in Teslas
- $\vec{v} \times \vec{B}$ is a cross-product
  - The product will be perpendicular to $\vec{v}$, $\vec{B}$.
  - It will be maximum when $\vec{v}$, $\vec{B}$ are perpendicular. Zero will be parallel.
  - Has magnitude $|\vec{v} \times \vec{B}| = vB \sin\theta$, with θ being the angle between $\vec{v}$ and $\vec{B}$.
  - Points in direction given by right-hand-rule 2.

2.1 Right Hand Rule #2

1. Use your right hand (dummy!)
2. Requires a flat hand
3. Thumb in direction of $\vec{v}$
4. Fingers pointing towards $\vec{B}$
5. Palm faces directon of $\vec{F}$
6. Switch direction for negative charges

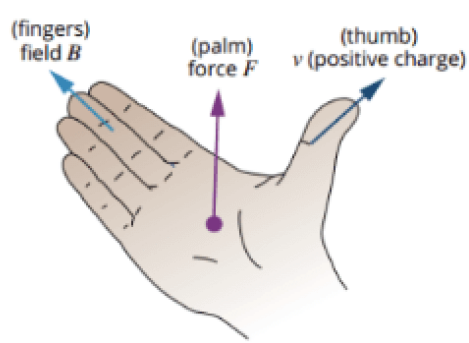

Figure 3: Second right-hand rule

Note that $\vec{v}$ and $\vec{B}$ don't need to be perpendicular.

2.2 Units

$$N = C\frac{m}{s}T$$

$$T = \frac{Ns}{cm}$$

3 Cross Products

$\vec{v} \times \vec{B} =$

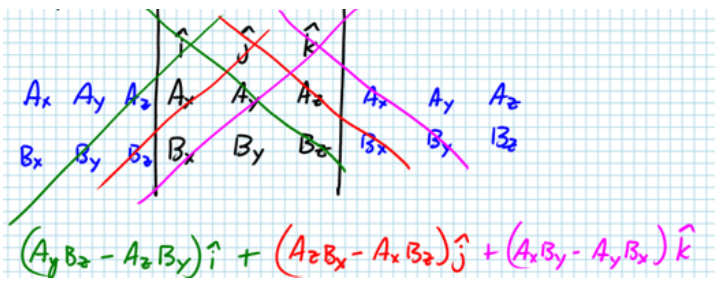

Figure 4: Cross product magnetism

- $\hat{i}$ or X:

$$(A_yB_z - A_zB_y)\hat{i}$$

- $\hat{j}$ or Y:

$$(A_zB_x - A_xB_z)\hat{j}$$

- $\hat{k}$ or Z:

$$(A_xB_y - A_yB_x)\hat{k}$$

3.1 Example Task

$q = -2c$, $\vec{v} = 3\hat{j}\frac{m}{s}$, $\vec{B} = 4\hat{h}T$, $\vec{F} =$? Using the cross-product rule:

$$(-2)(3 * 4)\hat{i} = -24\hat{i}$$

4 Force on a current-carrying wire

$$\vec{F} = q\vec{v} \times \vec{B}$$

And, since the units of $q * v$ are $\frac{Cm}{s}$, or $\frac{C}{s}m$,

$$\vec{F} = I\vec{l} \times \vec{B}$$

Where $\vec{l}$ is the length of the wire.

Note that thumb rule #2 works here, with thumb as current.